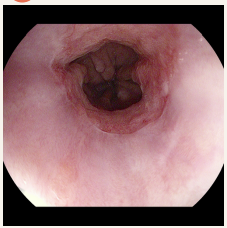
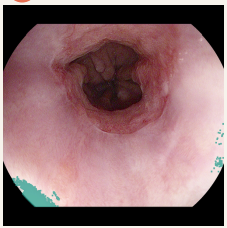
A Image preparation

1 White light frame



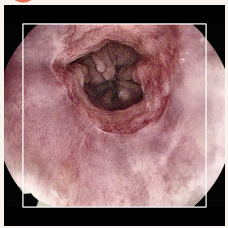
sample frame at the junction, overlay text removed

2 Specular highlights



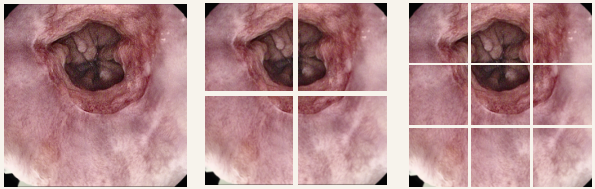
pixels with grey level > 240 (0.9% here, shown in green)

3 Inpaint, CLAHE, crop



Telea inpainting; CLAHE on LAB lightness (clip 2.0, 8×8); centered square

4 Fourteen multiscale views ($K = 1 + 4 + 9$)

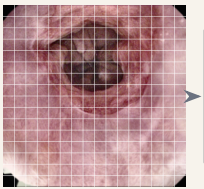


global, 224×224 | 2×2 quadrants | 3×3 tiles

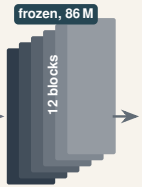
every view is resized to 224×224 before encoding

B Representation, ensemble and decision

5 Frozen DINOv2 ViT-B/14



16×16 patches of 14×14 pixels per view



[CLS] token

6 View embeddings

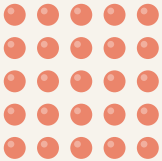
14 views × 768



(48 of 768 dimensions shown)

projection 768→256, LayerNorm, GELU

7 Mean pooling, 25 members



5 seeds × 5 folds grouped by source image; EMA weights

pooling chosen out of fold from five aggregators (AUPRC 0.623)

$$\mathbf{z} = \frac{1}{K} \sum_k \Phi(\mathbf{x}_k)$$

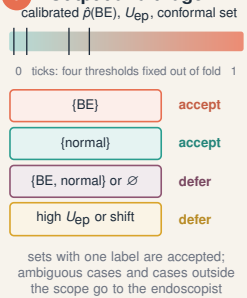
8 Calibration



ECE 0.080 → 0.044

50 Monte Carlo dropout passes per member

9 Output and triage



sets with one label are accepted; ambiguous cases and cases outside the scope go to the endoscopist



Where attention concentrates. Measured attention of the TransMIL comparator on BE test images, overlaid on the 3×3 tiles of this frame (darker coral means a higher share; center tile 1.42× uniform).

Out of fold only. The aggregator, calibrator, four thresholds and conformal quantiles are fitted on development predictions; test and external images never influence them.